

Hoa Huynh

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ABOUT

I am a Computer Vision Engineer with over 5 years of experience developing and deploying real-time deep learning systems on edge devices for automotive safety and surveillance. I am passionate about designing 2D/3D perception deep learning models, optimizing inference pipelines, integrating multi-sensor data, and collaborating with cross-functional teams to deliver practical AI solutions that improve human well-being. Additionally, I continuously seek out cutting-edge technologies, enhance my skills, and grow alongside the amazing people around me.

SKILLS

Programming Languages: C, C++, Python, C#, Java, JavaScript, MATLAB

Frameworks: Open3D, PCL, OpenCV, PyTorch, TensorFlow, Darknet, ONNX, TensorRT, TFLite, TIOVX, DeepStream, FFmpeg, Qt

MLOps & Tools: Git, SVN, Docker, TensorBoard, DVC, Weights & Biases, MLflow, CMake

SoCs: NVIDIA Jetson Nano/Xavier NX, Texas Instruments SK-TDA4VM/TDA4AEN, Samsung Exynos V920

Sensors: FLIR Thermal Camera, NIR Camera, Intel RealSense, LUCID Helios2 ToF Camera, Ouster LiDAR

Databases: Microsoft SQL Server, MySQL, SQLite, PostgreSQL, MongoDB

Operating Systems: Windows, Linux/Ubuntu, macOS, ROS

Languages: English (professional), Korean (elementary), Vietnamese (native)

EXPERIENCE

DeltaX

Seoul, South Korea

Development Team Lead & AI Researcher

Jun. 2022 – Present

- Since Dec. 2023, led a Research & Development team focused on building the In-Cabin Monitoring System (ICMS), overseeing technical roadmap definition, task allocation, weekly progress reviews, milestone management, and partner and client demonstrations:
 - * The ICMS is designed to comply with Euro NCAP standards and incorporates DeltaX's proprietary innovations to enhance vehicle safety through continuous monitoring and assessment of driver and passenger states.
 - * The solution uses a single RGB/IR fisheye camera, a C/C++ inference pipeline on a SoC device, and a web-based UI (Node.js + React) on a PC for real-time visualization. The inference pipeline integrates up to 8 deep learning models for object detection, 2D/3D body keypoints, facial landmarks, gaze estimation, and image classification, and delivers an average end-to-end frame rate of 14 FPS.
 - * Successfully delivered 4 proof-of-concept (PoC) projects for 2 key Korean automotive vendors within aggressive 3 – 6 month timelines: 42dot (Phase 1: Jan. – Jun. 2024; Phase 2: Aug. – Oct. 2024) and Pyeong Hwa Automotive (Phase 1: May 2024 – Jan. 2025; Phase 2: Jul. – Oct. 2025).
- Customized PyTorch models and optimized Python/C++ integration pipelines for SoC deployments.
- Deployed on TI SK-TDA4VM/TDA4AEN using Edge AI + TIOVX frameworks, and on Samsung Exynos V920 using an ONNX Runtime-based framework.
- Designed lightweight 2D/3D perception models and algorithms (facial analysis, semantic segmentation, monocular depth estimation, pseudo-LiDAR, visual SLAM, real-time multi-camera stitching, and head pose estimation) using RGB/IR pinhole and fisheye camera inputs with LiDAR/RGB-D supervision for applications including human localization for safety monitoring, 3D display technology, and advanced driver-assistance systems (ADAS).
- Built and maintained utilities for multi-sensor calibration, data collection, and analysis using ROS, MATLAB, and Python.

- Developed a dataset preprocessing algorithm inspired by “SAHI: Slicing Aided Hyper Inference” to improve recall on small objects for YOLOv8-p2/YOLOv9n models while maintaining real-time inference latency.

Smartdeep

Seoul, South Korea

Research Engineer

May 2021 – May 2022

- Built a multi-camera surveillance system aligned with KISA Intelligent CCTV Certification using C++ (OpenCV + FFmpeg) and a C# WPF GUI. The system supports up to 16 CCTV streams simultaneously and can detect, record, and search predefined events by analyzing moving or abnormal objects.
- Contributed to the full lifecycle of Darknet YOLOv4-based object detection models by curating datasets and optimizing the training and validation pipeline.

Ubay Solution

Seoul, South Korea

AI Research Intern

Dec. 2020 – Apr. 2021

- Developed an optical character recognition (OCR) system for deployment at Busan Port, enabling automated detection and recognition of truck license plate numbers and container codes.
- Evaluated the performance and reliability of object detection frameworks (SSD, YOLOv4, Faster R-CNN, etc.) on a custom dataset.
- Applied data augmentation and web crawling techniques to increase dataset size and quality.
- Implemented a DeepStream + TensorRT pipeline on NVIDIA Jetson Nano and Jetson Xavier NX for real-time inference.

ANDA Lab – Soongsil University

Seoul, South Korea

Research Assistant

Sep. 2018 – Aug. 2020

- Conducted research and experimental analysis using Python, OpenCV, and MATLAB to diagnose issues in a vehicle localization system centered on taillight detection and processing.
- Composed scientific papers proposing solutions to address the identified issues.

Global Cybersoft JSC

Ho Chi Minh City, Vietnam

Software Engineer Trainee

Mar. 2017 – Aug. 2017

- Developed C++ Microsoft Foundation Class (MFC) plugins for SCREEN Holdings Co.’s industrial printer system (Japan).
- Developed Adobe Illustrator plugins using the Adobe Illustrator SDK, C++, and JavaScript for an advertising agency (Japan).
- Improved code quality and team collaboration by participating in Jira-based peer code reviews.
- Gained hands-on experience in an agile software development environment.

EDUCATION

Soongsil University

Seoul, South Korea

Master of Engineering, Information Communication

Sep. 2018 – Aug. 2020

- **GPA:** 4.01/4.5
- **Thesis:** LED Taillight Matching Algorithm for Nighttime Vehicle Positioning
- **Ref:** oasis.ssu.ac.kr/search/i-discovery/edsker:edsker.000004851619?type=eds (*English*)

University of Information Technology – Vietnam National University

Ho Chi Minh City, Vietnam

Bachelor of Engineering, Software Engineering

Aug. 2013 – Jun. 2018

- **GPA:** 8.6/10 (Salutatorian Honors)
- **Thesis:** Coronary Arteries Segmentation from Coronary Angiogram
- **Ref:** drive.google.com/file/d/1RylFqRy-xywZG1zEWEX2l3B49T7XbFvf/view (*Vietnamese*)

PUBLICATIONS

(*I*: International Journal, *W*: International Workshop, *K*: Korean Domestic Journal/Conference)

K. Uddin S M Nadim et al. (including **Huynh Thai Hoa**), “A Railway Framework for Person (Worker) Localization Based on Depth Estimation and GPS”, *The Korean Institute of Defense Technology Fall Conference* (2024). Ref: kidet.or.kr/index.php?page=view&pg=&idx=1259&hCode=BOARD&bo_idx=1&sfl=&stx=

W. Jamie Spencer et al. (including **Huynh Thai Hoa**), “The Second Monocular Depth Estimation Challenge”, *Conference on Computer Vision and Pattern Recognition Workshops* (2023). Ref: arxiv.org/abs/2304.07051

I. Thai-Hoa Huynh, Myungsik Yoo, “Taillight Matching and Pairing Algorithm for Stereo Vision-based Nighttime Vehicle Positioning”, *MDPI Applied Sciences* (2020). Ref: mdpi.com/841776

K. Thai-Hoa Huynh, Myungsik Yoo, “Nighttime LED Taillight Detection Method with Optical Camera Communication System”, *Journal of Korean Institute of Communications and Information Sciences* (2020). Ref: dbpia.co.kr/journal/articleDetail?nodeId=NODE10440000

I. Thai-Hoa Huynh, Tuan-Anh Pham, Myungsik Yoo, “Detection Algorithm for Overlapping LEDs in Vehicular Visible Light Communication System”, *IEEE Access* (2019). Ref: ieeexplore.ieee.org/document/8792184

SELECTED CERTIFICATIONS

Hugging Face:

- Fundamentals of Large Language Models (2025)
- AI Agents Fundamentals (2025)
- Fundamentals of Group Relative Policy Optimization (2025)

Coursera:

- Introduction to Machine Learning in Production (2024)

HackerRank:

- Problem Solving – Intermediate (2024)

Kaggle:

- Computer Vision (2022)
- Feature Engineering (2022)
- Time Series (2022)

(Full certification list available on portfolio website: huynhthaihoa.github.io/#certifications)

REFERENCES

Myungsik Yoo: Master’s Thesis Advisor

- **Current Position:** Professor and Lab Director, ANDA Lab - Soongsil University (South Korea)
- **Email:** myoo@ssu.ac.kr

Trong-Hop Do: Former Postdoctoral Researcher at ANDA Lab

- **Current Position:** Vice Dean of the Faculty of Software Engineering, UIT - VNUHCM (Vietnam)
- **Email:** hopdt@uit.edu.vn

Sayed Nadim: Former Group Leader at DeltaX

- **Current Position:** Senior AI/ML Researcher, DEEP IN SIGHT Co., Ltd. (South Korea)
- **Email:** smnadimuddin@gmail.com

Vuong T. Pham: Bachelor’s Thesis Advisor

- **Current Position:** Senior Lecturer, Saigon University (Vietnam)
- **Email:** ptvuong@gmail.com